

The a.c. and d.c. Josephson effects in a Bose–Einstein condensate

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The alternating- and direct-current (a.c. and d.c.) Josephson effects were first discovered in a system of two superconductors, the macroscopic wavefunctions of which are weakly coupled via a tunnelling barrier^{1,2}. In the a.c. Josephson effect^{1–7}, a constant chemical potential difference (voltage) is applied, which causes an oscillating current to flow through the barrier. Because the frequency is proportional to the chemical potential difference only, the a.c. Josephson effect serves as a voltage standard². In the d.c. Josephson effect, a small constant current is applied, resulting in a constant supercurrent flowing through the barrier^{4,5,8}. In a sense, the particles do not ‘feel’ the presence of the tall tunnelling barrier, and flow freely through it with no driving potential. Bose–Einstein condensates should also support Josephson effects⁹; however, while plasma oscillations have been seen¹⁰ in a single Bose–Einstein condensate Josephson junction, the a.c. Josephson effect remains elusive. Here we observe the a.c. and d.c. Josephson effects in a single Bose–Einstein condensate Josephson junction. The d.c. Josephson effect has been observed previously only in superconducting systems¹¹; in our study, it is evident when we measure the chemical potential–current relation of the Bose–Einstein condensate Josephson junction^{4,11}. Our system constitutes a trapped-atom interferometer^{12,13} with continuous readout¹⁴, which operates on the basis of the a.c. Josephson effect. In addition, the measured chemical potential–current relation shows that the device is suitable for use as an analogue of the superconducting quantum interference device, which would sense rotation^{2,15}.

We can understand all of the effects reported here from the following three nonlinear equations describing the model of two Bose–Einstein condensates (BECs)⁷, which are weakly coupled in a double-well potential, in the presence of dissipation due to thermal atoms (see Supplementary Information). These equations are derived from the Gross–Pitaevskii equation (GPE)¹⁶, with the exception of the second term of equation (1), which represents the effect of non-zero temperature. The atomic current $\dot{\eta}$ flowing across the barrier is given by¹⁷:

$$\dot{\eta} = \omega_J \sin \phi - G \Delta\mu \quad (1)$$

where $\eta \equiv (N_1 - N_2)/N$ is the relative population between the two condensates with $N \equiv N_1 + N_2$, $\phi \equiv \phi_1 - \phi_2$ is the relative phase between the two condensates, $N\hbar\omega_J/2$ is the Josephson coupling energy, and G is the conductance due to thermal atoms^{5,17,18}. The chemical potential difference $\Delta\mu \equiv \mu_1 - \mu_2$ is given by⁴:

$$\Delta\mu = \hbar\omega_C (\eta - \eta_{\text{equil}}) \quad (2)$$

where $2\hbar\omega_C/N$ is the capacitive energy, and η_{equil} is the equilibrium value of η , which is non-zero for a double well with different volumes on the two sides. Equation (2) expresses the fact that a number imbalance is associated with a chemical potential difference, due to

repulsive interactions. The phase evolution is given by¹:

$$\dot{\phi} = -\frac{\Delta\mu}{\hbar} \quad (3)$$

Equation (3) is valid for any two well-separated condensates ($\omega_J \ll \omega_C$), whether they are coupled by tunnelling or not^{4,19}. This phase evolution has been observed^{12,13} in the Fock regime of completely uncoupled condensates ($\omega_C/\omega_J \gg (N/2)^2$), as well as for condensates in the Josephson regime ($1 \ll \omega_C/\omega_J \ll (N/2)^2$), for the case of $\omega_C/\omega_J \approx 15$ (ref. 10). The Josephson regime characterizes the current experiment, as well as the superconducting case¹⁹. Equations (1)–(3) describe each of the three nonlinear systems shown in Fig. 1 (ref. 2). These systems are analogous to the BEC Josephson junction, and will provide an intuitive description of the effects observed.

We will now discuss the various solutions of equations (1)–(3), beginning with the a.c. Josephson effect. Consider equations (1) and (3), with a constant $\Delta\mu$ applied. Then $\dot{\phi}$ is constant, and for $G = 0$, the current becomes $\dot{\eta} = -\omega_J \sin(\frac{\Delta\mu}{\hbar} t)$ (refs 2, 3). Thus, the current oscillates in time with frequency $\Delta\mu/\hbar$, as indicated by the solid line of Fig. 2a. This a.c. Josephson effect relies on the sinusoidal nature of equation (1), which results from the weak coupling between the two macroscopic wavefunctions²⁰. In contrast, in a bulk macroscopic quantum system, the current is proportional to the gradient of the phase, so equation (1) would be replaced by $\dot{\eta} \propto \phi$. Such a system would show no oscillating current because ϕ increases at a constant rate. As shown in Fig. 1a, the a.c. Josephson effect corresponds to the rigid pendulum executing complete rotations, with an angle that always increases². The period of rotation of the pendulum corresponds to the period of the current oscillations, $(\Delta\mu/\hbar)^{-1}$.

The a.c. Josephson effect is very different from the plasma oscillations indicated in Fig. 1a, for which the rigid pendulum oscillates resonantly about zero with $-\pi < \phi < \pi$ (ref. 21). The oscillations observed in ref. 10 were plasma oscillations, as were the oscillations in the lattice of ref. 22. These plasma oscillations exist even for small-angle oscillations, for which equation (1) reduces to $\dot{\eta} \propto \phi$, as in a bulk macroscopic quantum system. The a.c. Josephson effect, in contrast, is a striking demonstration of the sinusoidal nature of equation (1), and has no analogue in a bulk system.

The a.c. Josephson effect has been seen in superfluid ³He (ref. 23), as well as in superconductors. In BEC, this effect has been seen in systems where the chemical potential difference $\Delta\mu$ is produced by an asymmetric potential rather than by a population imbalance. This includes a lattice of Josephson junctions²⁴, as well as an internal a.c. Josephson effect²⁵. In the single BEC Josephson junction reported here, $\Delta\mu$ is produced by a population imbalance, as in the superconducting case.

Our ultrahigh-resolution BEC apparatus applies optical potentials through an aspheric lens with a numerical aperture of 0.5, as shown in Fig. 3a. The imaging is performed through this same lens. The

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observed resolution for creating potentials as well as imaging is $1.2\ \mu\text{m}$. In the current experiment, the apparatus is used to create a double-well Josephson junction, as shown in Fig. 3b. The tunnelling barrier of the double well is a highly elongated laser beam, with a $1/e^2$ diameter of $1.4\ \mu\text{m}$ along the tunnelling axis and $400\ \mu\text{m}$ along the long axis. The laser's wavelength of $774\ \text{nm}$ is blue-detuned from resonance ($780\ \text{nm}$). The barrier divides the axially symmetric, harmonic magnetic trap (measured trap frequencies of $\omega_r/2\pi = 224\ \text{Hz}$ and $\omega_z/2\pi = 26\ \text{Hz}$) parallel to its long axis. The position of the centre of the harmonic trap relative to the barrier is controlled by an additional magnetic coil. After transporting the ^{87}Rb atoms in the ($F = 2$, $m_F = 2$) state to the high-resolution region via a moving quadrupole trap²⁶, evaporation is performed in the double well, resulting in $N = 10^5$ atoms in the two condensates.

To observe the a.c. Josephson effect, we create a chemical potential difference $\Delta\mu$ between the two clouds, and measure the frequency of the oscillations which accompany the almost constant $\Delta\mu$. First, BEC is reached by evaporating into an asymmetric double well potential, as shown in the inset of Fig. 2a. The harmonic trap is then shifted over 5 to 10 ms to form a symmetric double well, but with a chemical potential difference, as shown in the inset. After the shift, twelve non-destructive phase contrast images are taken of the BEC Josephson junction, giving $\Delta\mu$ and $\eta(t)$, as shown in Fig. 4a–c. The oscillation frequencies of the two curves of Fig. 4c are seen to be proportional to $\Delta\mu$. To find the frequency of oscillation ω , the

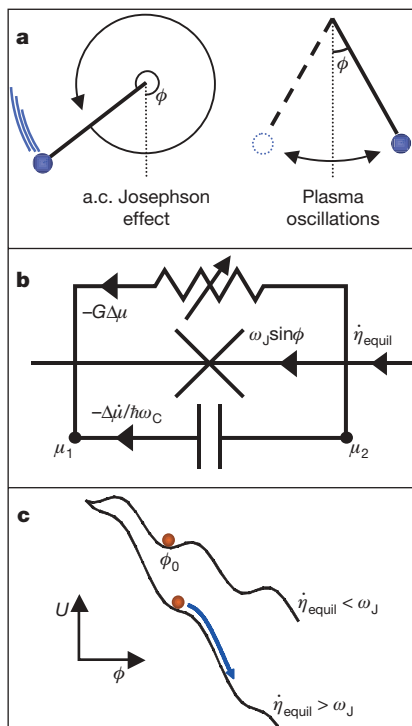


Figure 1 | Nonlinear systems analogous to the BEC Josephson junction. Each of these systems represents equations (1)–(3). The damping time of each of these systems is of the order of $(\hbar G \omega_C)^{-1}$. The conductance G is varied by adjusting the fraction of thermal atoms. **a**, The rigid pendulum. The case of zero current bias ($\dot{\eta}_{\text{equil}} = 0$) is shown. The relative phase ϕ is represented by the angle of the rigid pendulum, and the population imbalance η is proportional to the speed $\dot{\phi}$. **b**, Superconducting circuit, analogous to the resistively shunted Josephson junction model. The $\times$ -shape represents the Josephson junction. The current through each of the branches is indicated on the figure. The constant applied current bias is given by $\dot{\eta}_{\text{equil}}$. **c**, A particle moving in a washboard potential. By computing the energy $\int \Delta\mu dN$ for each of the currents in **b**, we find that the system is equivalent to a particle of mass $N\hbar/2\omega_C$ moving in a washboard potential given by $U = (\hbar N/2) [\omega_J(1 - \cos \phi) - \dot{\eta}_{\text{equil}} \phi]$. The tilt of the washboard potential is due to the current bias $\dot{\eta}_{\text{equil}}$.

Fourier transform of $\eta(t)$ is computed, and the peak of the Fourier transform is taken as ω (ref. 23). Figure 2a shows the resulting measurement of the a.c. Josephson effect in a BEC. The measured frequency ϕ of the atomic current gives the chemical potential difference via equation (3).

Although the a.c. Josephson effect has not been observed previously in a BEC Josephson junction, a related effect has been seen, which we now describe. In the absence of dissipation ($G = 0$), the speed of the freely rotating rigid pendulum averaged over a complete rotation never decreases, which implies that the population imbalance in the a.c. Josephson effect, averaged over a complete period, never decreases¹⁹, as indicated by the upper, blue solid curve of Fig. 2b. In other words, in the absence of dissipation, no net current flows in the a.c. Josephson effect, only an alternating current. This non-equilibrium associated with the underlying a.c. Josephson effect is referred to as “macroscopic quantum self-trapping” (MQST)^{7,19}, which was observed in a BEC¹⁰. In Fig. 1b, the freely rotating rigid pendulum is equivalent to $\dot{\eta}_{\text{equil}} = 0$, with a charged capacitor. MQST reflects the fact that owing to the a.c. Josephson effect, the capacitor will never discharge.

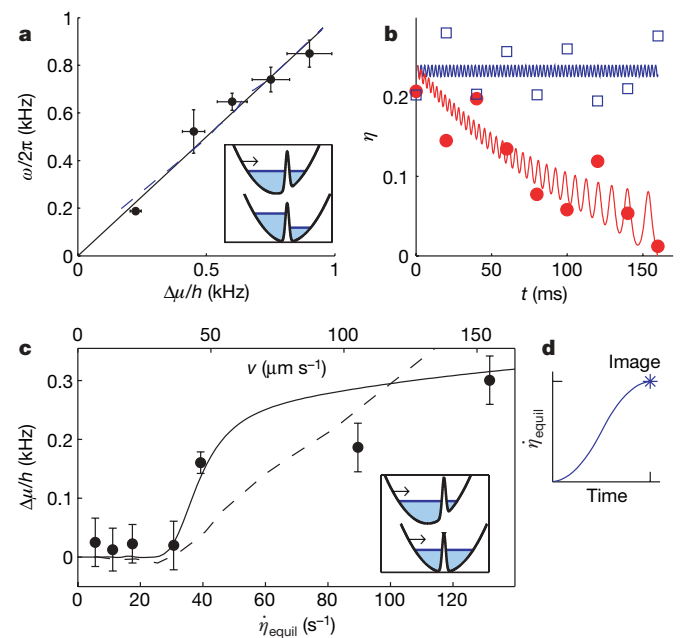


Figure 2 | Observation of the a.c. and d.c. Josephson effects. **a**, The a.c. Josephson effect. The solid line shows $\omega/2\pi = \Delta\mu/h$, with no free parameters. The dashed curve is a full simulation of the GPE. Error bars show the standard error of the mean, for between two and twelve repetitions of the experiment. The inset shows the creation of $\Delta\mu$. **b**, The decay of the MQST. The measured blue squares show a long damping time, with a thermal fraction of roughly 5% ($T \approx 0.3T_c$) (ref. 16). The increased damping seen for the red circles is with a thermal fraction of roughly 20% ($T \approx 0.5T_c$), obtained by increasing the final evaporation frequency by 2.5 kHz. The solid curves are numerical solutions of equations (1)–(3). In the decaying red curve, $(\hbar G \omega_C)^{-1}$ is adjusted to 0.1 s. **c**, The $\Delta\mu$ – I relation and the d.c. Josephson effect. Filled circles are the measured values, which show the d.c. Josephson effect for $\dot{\eta}_{\text{equil}} < 30\ \text{s}^{-1}$. Three of the points are averages of two measurements. For these points the error bars indicate the standard error of the mean. The average of these error bars, multiplied by $\sqrt{2}$, gives the error bars shown for the other points. The solid line is a numerical solution of equations (1)–(3). The dashed curve is a GPE simulation, in which the barrier height is adjusted to a value of $(2.5\ \text{kHz}) \times \hbar$. The inset shows the application of $\dot{\eta}_{\text{equil}}$. **d**, Before imaging each point in **c**, $\dot{\eta}_{\text{equil}}$ is increased slowly, to prevent plasma oscillations in the potential of Fig. 1c. Any plasma oscillations around ϕ_0 would allow the system to leave the supercurrent state, and enter the finite $\Delta\mu$ state, even for $\dot{\eta}_{\text{equil}} < \omega_J$. With $\dot{\eta}_{\text{equil}}$ at its final value, which occurs at the moment that the potential well is symmetric, we image the atoms and determine $\Delta\mu$ by the same technique shown in Fig. 4.

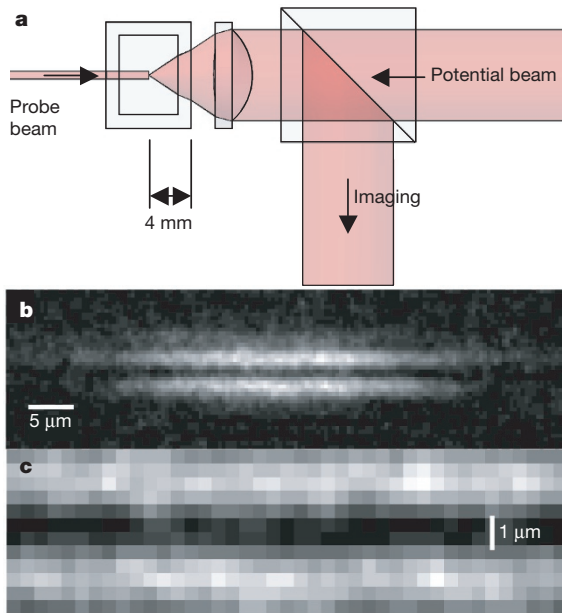


Figure 3 | Creating and imaging our BEC Josephson junction. **a**, The application of high-resolution potentials, and imaging with the same resolution. Almost arbitrary potentials can be formed by engineering the potential beam. The left rectangular region is the cross-section of the vacuum cell. **b**, *In situ* image of our BEC Josephson junction. The light regions are the two BECs. The image is obtained with destructive phase contrast imaging, employing small detuning to improve the signal to noise ratio. **c**, An enlargement of the narrow tunnelling region of the wavefunction. Each pixel is $0.46 \mu\text{m}$. Gravity is directed downwards in all panels.

However, we observe that if damping due to the thermal atoms is non-zero ($G > 0$), the speed of the rigid pendulum will slowly decrease, corresponding to decay of the MQST^{5,18,27}, as indicated

by the lower, red solid curve of Fig. 2b. This decay occurs on a time-scale determined by the fraction of thermal atoms, as indicated by the two sets of measured points in Fig. 2b. This timescale is much longer than the time between images in a measurement such as Fig. 4a, which varies between 0.3 and 1.7 ms. The decay of the MQST shows that the thermal cloud significantly modifies the behaviour of the two condensates. This can be understood from equations (1)–(3). The second term of equation (1) implies that $\Delta\mu$, which is mostly due to the condensate atoms, drives the motion of the thermal atoms^{5,17,18}. Furthermore, by equations (2) and (3), the resulting population change in the thermal atoms contributes to $\Delta\mu$ and drives the phase evolution of the condensate.

We now turn to the d.c. Josephson effect^{1,4,5,8,11}, which can be understood in terms of the washboard potential shown in Fig. 1c. As long as the potential is not tilted too much by the applied current $\dot{\eta}_{\text{equil}}$, the system can sit in a minimum with phase ϕ_0 given by $\dot{\eta}_{\text{equil}} = \omega_J \sin \phi_0$. Because $\dot{\phi}$ is zero, $\Delta\mu$ is zero by equation (3). Thus, a constant supercurrent $\dot{\eta} = \dot{\eta}_{\text{equil}}$ flows through the barrier with no driving potential difference^{4,11,21} (the d.c. Josephson effect), as indicated by the solid black curve of Fig. 2c for small $\dot{\eta}_{\text{equil}}$. On the other hand, when $\dot{\eta}_{\text{equil}}$ exceeds the critical current ω_J , some current must flow through the resistive or capacitive paths shown in Fig. 1b. These currents result in a non-zero $\Delta\mu$. The relation between the applied current bias $\dot{\eta}_{\text{equil}}$ and the resulting $\Delta\mu$ (the $\Delta\mu$ – I relation) thus has both a supercurrent state and a finite $\Delta\mu$ state, with a transition between the states at $\dot{\eta}_{\text{equil}} = \omega_J$, as indicated by the solid curve of Fig. 2c. The two states are also represented by the two curves of Fig. 1c.

To measure the $\Delta\mu$ – I relation, we essentially perform the experiment suggested in ref. 4. A bias current $\dot{\eta}_{\text{equil}}$ is applied by moving the harmonic potential at a speed v relative to the barrier, by means of the additional magnetic coil, as shown in the inset of Fig. 2c. The total distance moved is $0.7 \mu\text{m}$. Figure 2c shows the resulting measurement of the $\Delta\mu$ – I relation of the BEC Josephson junction.

The supercurrent state is seen in Fig. 2c for $\dot{\eta}_{\text{equil}} < 30 \text{ s}^{-1}$ ($v < 40 \mu\text{m s}^{-1}$). This state represents the d.c. Josephson effect^{4,5,8}. The thus measured critical current of 30 s^{-1} is ω_J from equation (1). By comparing $\Delta\mu$ and η , and using equation (2), we find a value

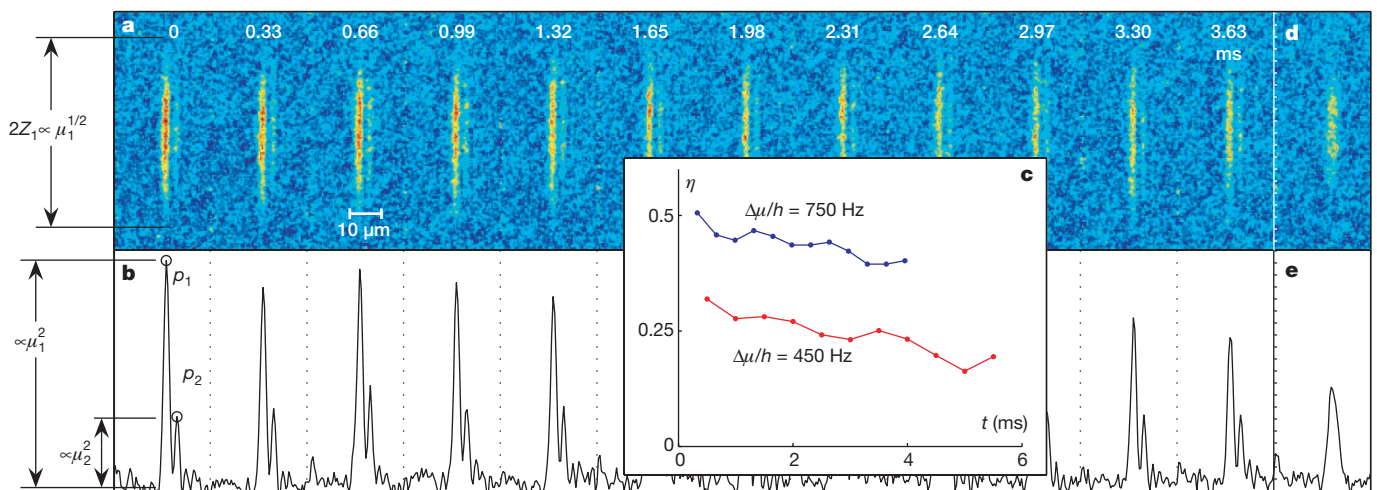


Figure 4 | Time evolution of a BEC Josephson junction. **a**, Twelve *in situ* images of the same BEC Josephson junction, obtained by non-destructive phase contrast imaging. To obtain an absolute, quantitative value μ_1 for the chemical potential of one of the clouds, its length $2Z_1$ is measured directly from a fit to the image, as shown. This gives the chemical potential from the relation $\mu_1 = m\omega_z^2 Z_1^2 / 2$, where m is the atomic mass. The number of atoms N_1 or N_2 is computed by integrating over each cloud, giving $\eta(t)$ for each of the twelve images. **b**, The image integrated in the z -direction. The height of the peaks p_1 and p_2 are proportional to μ_1^2 and μ_2^2 , respectively, in the Thomas–Fermi approximation¹⁶. The chemical potential difference is thus

given by $\Delta\mu = \mu_1 - \mu_2$ where $\mu_2 = \mu_1 \sqrt{p_2/p_1}$. **c**, The phase evolution of the BEC Josephson junction. The top, blue curve shows $\eta(t)$, when averaged with seven other repetitions of the experiment. For comparison, the lower, red curve shows a smaller oscillation frequency corresponding to a smaller value of $\Delta\mu$. The slope of the curves is due to imaging losses. Panels **d** and **e** show the BEC in the harmonic trap. After the images shown in **a** and **b** are obtained, the barrier is slowly decreased to zero. The image of the resulting BEC shows the profile of the harmonic trap, via the Thomas–Fermi approximation¹⁶. We use this profile to verify the alignment of the centre of the harmonic trap with the barrier.

of $\omega_c = 9,000 \text{ s}^{-1}$. Because $\omega_c/\omega_J = 300$, the system is in the Josephson regime: $1 \ll \omega_c/\omega_J \ll (N/2)^2$. As the current increases beyond the critical current in Fig. 2c, the system enters the finite $\Delta\mu$ state. This transition is expected to occur via the disappearance of the minimum in Fig. 1c rather than via fluctuations, because the height of the effective potential barrier of Fig. 1c is on the order of $\hbar N\omega_J/k_B \approx 20 \text{ } \mu\text{K}$. This is orders of magnitude larger than the actual temperature ($<100 \text{ nK}$), or the effective temperature of quantum fluctuations, which is of the order of the plasma oscillation frequency in the washboard potential $\hbar\sqrt{\omega_c\omega_J}/k_B \approx 4 \text{ nK}$ (refs 21, 28).

Our BEC Josephson junction can be considered to be a trapped-atom interferometer^{12–14}. Like the system of ref. 14, our system gives a continuous measurement of the relative phase between the two coherent clouds of atoms, as shown in Fig. 4c. However, the operation of our interferometer is based on the a.c. Josephson effect, rather than on the non-tunnelling technique of ref. 14. We verify the functioning of the interferometer by measuring various interaction energies, as shown in Fig. 2a. Our interferometer could also be used to measure the time evolution of applied optical or magnetic potentials¹², or to measure a time-varying scattering length. If our interferometer were to be used for a relatively long measurement, it would be essential that $\Delta\mu$ did not equilibrate through the tunnelling barrier. Fortunately, this occurs naturally, owing to the MQST, which occurs for sufficiently low temperatures.

Furthermore, two BEC Josephson junctions could be combined in a loop, to form the analogue of a superconducting quantum interference device (SQUID), which would sense rotation. Such a BEC SQUID would require a macroscopic wavefunction around the entire loop, because it relies on the quantization of phase around the loop. The BEC SQUID would have a $\Delta\mu$ – I relation as in Fig. 2c, except that the critical current $\dot{\eta}_c$ would vary with the rate of rotation. The sensitivity to rotation would be proportional to $\delta\dot{\eta}_c/\dot{\eta}_c$, where $\delta\dot{\eta}_c$ is the smallest measurable change in $\dot{\eta}_c$. It is interesting to note that the $\Delta\mu$ – I relation of Fig. 2c shows a rapid variation in $\Delta\mu$ for a small variation in $\dot{\eta}_{\text{equil}}$. Thus, by measuring $\Delta\mu$ in the BEC SQUID, one could measure the change in $\dot{\eta}_c$, and rotation, with high sensitivity^{2,15}. This is analogous to a superconducting SQUID's sensitivity to magnetic flux². A SQUID detector of rotation was achieved in superfluid ^3He (ref. 29), but the detection did not exploit the sensitivity gain of the $\Delta\mu$ – I relation. A BEC SQUID differs from a Sagnac-effect atom interferometer in that it requires a macroscopic wavefunction around the loop, and $2\delta\dot{\eta}_c/\dot{\eta}_c$ replaces the minimum measurable phase³⁰. In principle, owing to the $\Delta\mu$ – I relation, a BEC SQUID could have more sensitivity than a Sagnac atom interferometer of the same enclosed area.

Our BEC Josephson junction is the first application of our ultra-high-resolution BEC system, allowing for almost arbitrary potentials and imaging on a tunnelling length scale. We report the measurements of the a.c. and d.c. Josephson effects in a BEC, as well as the measurement of the $\Delta\mu$ – I relation. Our BEC Josephson junction can be considered to be an atom interferometer, in which the MQST has the important role of maintaining the chemical potential difference over time. The thermal cloud is seen to significantly alter the dynamics, in that it can cause the decay of the MQST. Furthermore, this system is seen to be suitable for forming the analogue of a SQUID detector, which would sense rotation. The work reported here represents the start of a new type of cold atom experimentation, where the tunnelling can be observed *in situ*, without opening the confining potential.

METHODS SUMMARY

The aspheric lens of Fig. 3a is of a type typically used for coupling a laser beam to a fibre. Specifically, it is a Thorlabs 350240-B, with a focal length of 8 mm. The laser for the potential beam is a Ti:sapphire ring laser (Coherent 899-21), pumped by a frequency-doubled Nd:YVO₄ laser (Coherent Verdi V-10). The long axis of the harmonic trap is oriented perpendicularly to the image of Fig. 3a.

In Fig. 4a, twelve images were obtained of the same BEC Josephson junction, where each image gives the number of atoms in each of the two weakly coupled BECs. Making these measurements of the number of atoms, with a precision limited by shot noise, does not randomize the relative phase. For the large number of atoms involved, the relative number and phase can be considered simultaneously well defined^{17,19}.

Creating Fig. 4b involved integrating the entire image into a small number of points, which greatly reduces noise. Therefore, we used Fig. 4b to find the ratio between the chemical potentials of the two clouds. The relation $p_{1,2} \propto \mu_{1,2}^2$ in Fig. 4b stems from the parabolic nature of the trap in two dimensions. To obtain an absolute calibration, however, we use the length of the cloud in Fig. 4a. We use the left cloud because it is longer and has more atoms, so we can measure its length more accurately. We find that this method minimizes the noise, while maintaining an absolute calibration of $\Delta\mu$.

To compute the lower horizontal axis from the upper horizontal axis in Fig. 2c, we need the ratio between $\dot{\eta}_{\text{equil}}$ and v , which is the ratio between η_{equil} and x . We compute this ratio $d\eta_{\text{equil}}/dx$ by computing the ground-state wavefunction by the GPE for a given x . We find $d\eta_{\text{equil}}/dx = 0.85 \text{ } \mu\text{m}^{-1}$. In Fig. 2d, $\dot{\eta}_{\text{equil}}$ is ramped up in a time varying between 8.5 and 200 ms. The total distance moved is always $0.7 \text{ } \mu\text{m}$, so $\dot{\eta}_{\text{equil}}$ and v are determined by the time of the ramp.

Full Methods and any associated references are available in the online version of the paper at www.nature.com/nature.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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METHODS

The harmonic magnetic trap in the high-resolution region is a QUIC-trap³¹, which is the coil configuration typically used in conjunction with a moving quadrupole trap²⁶. The quadrupole coils of the QUIC-trap are located above and below the vacuum cell in Fig. 3a. The vacuum cell is anti-reflection coated on the outside only.

In ref. 14, a trapped-atom interferometer with continuous measurement was achieved by a non-tunnelling technique. There, the measurement of the phase only perturbed the interferometer through the extraction of a very small sample of atoms. Here, the measurement only perturbs the interferometer through the slight scattering associated with phase-contrast imaging³².

The imaging process integrates the density along the y direction (perpendicular to the image plane), so Fig. 4b is integrated in both the y and z directions, for both of which the trap is parabolic. The Thomas–Fermi radii in both of these directions are proportional to $\sqrt{\mu_1}$ and $\sqrt{\mu_2}$ for clouds 1 and 2, respectively. This implies that the height of the peaks p_1 and p_2 are proportional to μ_1^2 and μ_2^2 , respectively. This is verified by a GPE simulation.

In the measurement of the $\Delta\mu$ – I relation, the position of the harmonic trap relative to the barrier is adjusted by applying a current I_c to an additional magnetic coil. To calibrate the upper horizontal axis of Fig. 2c, we apply a known current I_c through the coil, and observe the distance x over which the cloud shifts in the image. For a given current ramp $I_c(t)$, we thus know $v(t)$ of the upper axis, because $v(t) = \frac{dx}{dt} \frac{dI_c(t)}{dt}$.

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